The Supermax Trap:

An Analysis of Roster Construction and Winning efficiency in the NBA

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“*Basketball is my favorite sport; I like the way they dribble up and down the court”*

-Kurtis Blow, 1984

Introduction

The Economic Landscape of the NBA

In the modern National Basketball Association (NBA) each team is a multifaceted firm operating within a highly regulated and capital-intensive environment. With the implementation of the “Supermax” contract in the 2017 Collective Bargaining Agreement (CBA), the league introduced a new type of contract allowing teams to allocate up to 35% of the league salary cap to a single player. While the intent was to help small market teams retain high level talent, this policy created a new economic problem, “The Supermax Trap”. This research explores if such extreme levels of capital concentration is an efficient use of resources or detrimental to the team’s ability to succeed.

Resource Allocation Under Scarcity

At its core, this is a study of resource allocation under scarcity. Every NBA franchise operates under a Soft Cap with increasingly punitive “Aprons” as your spending progresses beyond the soft cap. This creates a de facto hard ceiling on spending for teams even though there is not a set limit. A General Managers primary task is to maximize output, defined as regular season wins, using this finite level of capital.

However, the labor market for NBA talent is not perfectly competitive. Contractual floors, ceilings, as well as exceptions dictated by the CBA, force teams into specific spending brackets that may not align with the marginal product of the athletes. In addition, certain teams

have brand or location-based benefits that incentivize players to accept lower pay. These market forces often force small market teams to overpay to attract or retain high level talent relative to their marginal production, creating an inefficient market.

The Winning Multiplier: Local Economic Effects

The importance of this research extends beyond the front office and onto the streets of the franchise home city. A winning NBA team acts as a positive externality for its local economy. In economies the "Winning Multiplier" effect dictates the team success enhancing the localized commerce.

- **Gate and Playoff Revenue:** an efficient roster leads to more home games in the post season maximizing the organizations yearly profits.
- **Local Macroeconomics:** Winning teams create an increased ticket demand which stimulates spending in the hospitality tourism, and service industries directly boosting local tax receipts.

Literature Review

The economic study of the NBA has evolved from a simple evaluation of raw spending to a complex analysis of capital concentration and organizational efficiency. Central to the current discourse is the work of (Cynkar 2025) in "Optimal Capital Allocation Decisions and Investments in the NBA" who identified a concave inverted U-shaped curve in a capital allocation that suggests diminishing marginal wins. (Cynkar 2025) suggested that the optimal allocation for the top player was 23.7% and to the top 2 players was 42.4%, providing a theoretical foundation for our Quadratic model.

Utilizing a different approach (Wang and Sarker 2025) in “The Effect of Basketball Analytics Investment on National Basketball Association (NBA) Team Performance” examine how off-court human capital impacts NBA team performance. They do this by establishing that analytics staff headcount is a statistically significant control variable. Finding that investment into analytic staff was a more efficient investment per dollar than on court capital.

While Wang and Sarker establish that analytics investment leads to a statistically significant increase in wins, our research narrows the scope to identify where that efficiency is realized. We hypothesize that the “Moneyball Effect” identified by (Wang and Sarker 2025) is most visible in a team’s ability to optimize its 5th starter (pct5) and aggregate depth (pct45). By integrating their findings on analytics headcount with our focus on institutional constraints (the Supermax), we can test if savvy front offices can utilize data to survive the Supermax Trap that (Cynkar 2025) identifies as a threat to roster balance.

Ultimately, this paper synthesizes these two perspectives, Cynkar’s (2025) warning about salary concentration and Wang and Sarker’s (2025) evidence that analytical human capital can alleviate those constraints. In our model we expand the analysis of salary concentration beyond the top 2 players to include the top 5 players. In addition, by including Injured Pct, Analytics Staff and lagged wins as controls, we provide a holistic view of the modern NBA franchise as a firm that must balance high-cost labor with high functioning management to maximize production.

Economic Framework

The NBA franchise as a Firm

In neoclassical economic theory a firm is an entity that transforms inputs into marketable output. For an NBA franchise, the output is defined as regular season success, specifically measured by total wins. While wins serve as a proxy for team quality, they also act as the primary driver for organizational revenue, additional playoff revenue, and brand equity.

The inputs for thus model are capital intensive. Unlike a standard manufacturing firm that balances labor and physical capital, an NBA team's primary input is the strategic allocation of a soft salary cap system across fifteen roster spots. For this analysis, we focus on the "Starting Five", or the top five earners for a team each year, as these players consume the vast majority of a team's financial resources.

Substitution Effects and Opportunity Costs of Depth

Economically, the decision to sign a Supermax player is not made in a vacuum, it carries a massive opportunity cost. In a salary cap system managed by aprons every dollar allocated to a primary star is a dollar stripped from the complementary inputs, players 2-5.

This paper explores a substitution effect between elite talent and roster depth. If the marginal product of a 5th starter (Role Player) is higher per dollar than the marginal product of the Star's final 5% of salary, the team is acting irrationally. Our model tests the hypothesis that Returns to scale in the NBA are not linear. By utilizing a Quadratic model, we can identify the "Efficiency Frontier" which is the specific distribution of wages that minimizes waste and maximizes win probability.

Diminishing Marginal Returns and the Usage Bottleneck

A cornerstone of this analysis is the Law of Diminishing returns. In most industries, adding more high-quality capital increases output, but at a decreasing rate. In basketball, this effect is driven by physical constraints, there is only one ball, 48 minutes per game time per position, and a finite number of possessions a game.

As a team moves from a "Max" player (25-30% of the cap) to a Supermax player (35%), the marginal win contribution often fails to keep pace with the marginal cost. This creates a Usage Bottleneck. If a team pays 35% for a player whose marginal product is capped by the presence of other high usage teammates, the team is essentially overpaying relative to the output. This phenomenon suggests that's the relationship between top tier spending (pct1) and wins is concave. Meaning there is a point where spending more on your top player actually decreases the team's ability to win by starving the rest of the roster of necessary funds to acquire talent.

The Supporting Cast's Convexity: Increasing Returns to Scale

A unique finding of this empirical research is that the production of wins does not follow a uniform path of diminishing returns across all inputs. Instead, the data for the 2nd and third highest paid roster spots (pct2 and pct3) suggests a convex relationship. This indicates a region of increasing returns to scale with more salary allocated to secondary stars leading to more wins at an increasing rate. This divergence suggest that roster construction is not a linear progression but rather a balancing of two distinct economic phenomena.

Wage Elasticity of Wins

Finally, we introduce the concept of Wage Elasticity. This measures of sensitive a team's win total are to a 1% change in salary for a specific roster spot. We anticipate that for the 5th roster spot (pct5), the elasticity will be significantly higher than expected. This suggests that depth is an undervalued asset in the NBA market. While Top-End Talent is frequently overvalued due to prestige and the marketing utility of superstars, which often diverges from their pure utility to producing wins.

Data Description

Institutional Constraints: Understanding Contract types in the NBA

The NBA has four different classifications for large scale contracts as defined in the CBA:

- **Standard Max Contract:** 25% of the NBA Salary Cap the year the contract is signed
- **Veteran Max Contract:** 30% of the NBA Salary Cap (7-9 years of experience)
  35% of the NBA Salary Cap (10+ Years of experience)
- **Rookie Supermax Contract:** Starts at 25% of the NBA Salary Cap converts to 30% if player makes All-NBA or MVP/DPOY in 4th year or 2 of 3 preceding years
- **Veteran Supermax Contact:** 35% of the NBA Salary Cap (7-9 years of experience) and player has earned MVP/DPOY or All-NBA Team

To represent the impact of these contracts four variables were created for our model:

- **vt_supermax:** represents any contract for a player with more than seven years of experience that commands a cap hold of 35%.
  - This variable represents both the 10+ year veteran max and the veteran supermax contracts; they were combined as they represent a similar economic burden.
- **rk_supermax:** represents the players that signed a rookie supermax extension, and it converted to the 30% cap hold.
- **Max:** represents any contract that is between 25% and 30% of the league cap.
- **AllStar:** represents if a team had a player selected to the All-Star team in the year being analyzed.
  - This is used as a proxy for talent as well to analyze if a star player who is up for a new contract is worth allocating significant salary to.

Percentage Variables: Normalization

Just as inflation makes comparing prices across time periods not a 1 to 1 comparison, the NBA's salary cap is adjusted every year depending on profits and TV deals. To combat this issue and provide more interpretable results key salary metrics have been normalized into parentages of the total salary expenditure for each team in a given year.

- **pct1-5:** represent the percentage of salary spending allocated to a team's top 5 highest paid players
- **pct45**: represents a consolidation of the 4th and 5th highest paid players to examine the effects of high-quality depth

- **pct1^2, pct2^2, and pct3^2:** created to analyze the nonlinear effect of top end salary spending

Control variables: Managing Statistical Noise

To ensure that observed relationship between salary allocation and winning are not biased by external factors, several control variables are included in the regression models. These variables clean the data by accounting for exogenous shocks and organizational quality that could otherwise lead to omitted variable bias

- **Injury Impact (Injured_pct):** represents the percentage of salary lost to injuries in a season. Calculated on a salary per game basis for games missed due to injury.
- **Team Continuity (Wins_t_minus_1):** Teams that were good in the previous year tend to be good in the subsequent year this variable introduces lagged wins to account for this.
- **Organizational Human Capital (Analytics_Staff):** this variable accounts for the headcount of analytics staff on an NBA team used as a proxy for the "Moneyball Effect" and talent identification.
- **Time (Time_Period):** this variable accounts for variation over time across the observed period of the 2017-18 NBA season to the 2025-26 NBA season.

Sourcing:

- **Contract, Record, and Salary Data:** These variables were sourced from Spotrac.com

- **Analytics Staff:** This variable was sourced from NBAstuffer.com and utilized the wayback machine to gather historical data closest to the start of each respective NBA season. Data was not available for the 2018-19 season, so values were imputed as the mean of the previous and subsequent season values.
- **AllStar:** This variable was sourced from RealGM.com

Empirical Models

This research consists of two separate models. One linear model that evaluates the relationship between wins and contract classification along with several controls to isolate the effect. Then a quadratic model that examines the ideal percentage ranges of salary allocated to roster spots 1 through 5. By using 2 models we can answer both whether Supermax contracts are inefficient, as well as how much a team should pay to maximize productivity.

The Linear Model: Controlling for Noise

$$\mathbf{Wins}_{it} = \beta_0 + \gamma \mathbf{Wins}_{it-1} + \beta_1 \mathbf{AllStars}_{it} + \beta_2 \mathbf{Max}_{it} + \beta_3 \mathbf{rk_superMax}_{it} + \beta_4 \mathbf{vt_supermax}_{it} + \beta_5 \mathbf{Injured_pct}_{it} + \beta_6 \mathbf{Analytics_Staff}_{it} + \beta_7 \mathbf{pct45}_{it} + \in_{it}$$

The Quadratic Model: Recipe for Success

$$\mathbf{Wins}_i = \beta_0 + \beta_1 \mathbf{pct1}_i + \beta_2 \mathbf{pct1}^2_i + \beta_3 \mathbf{pct2}_i + \beta_4 \mathbf{pct2}^2_i + \beta_5 \mathbf{pct3}_i + \beta_6 \mathbf{pct3}^2_i + \beta_7 \mathbf{pct4} + \beta_8 \mathbf{pct5} + \in_i$$

Expected Relationships

Positive:

- $\mathbf{Wins}_{it-1}$: Expected positive as teams that were good last year tend to stay good
- $\mathbf{AllStars}_{it}$: Expected positive as the more high-level player you have you tend to win more

- **Max$_{it}$**: Expected positive as they represent high level player without extreme salary allocation
- **Analytics_Staff$_{it}$**: Expected positive as high level of human capital leads to better talent identification
- **pct45$_{it}$**: Expected positive as quality depth leads to better team performance
- **pct4/5**: Expected positive relationship as higher investment into quality depth increases productivity

Negative:

- **rk_supermax$_{it}$** and **vt_supermax$_{it}$:** Expected negative due to concentration of salary into one player
- **Injured_pct$_{it}$**: Expected negative as more salary lost to injury the less availability you have of key players

Concave:

- **pct1:** overinvestment into one player is expected to have diminishing marginal returns

Convex:

- **pct2:** investment into a secondary star is expected to have increasing returns to scale as it represents commitment to high level team dynamics
- **pct3**: investment into a tertiary star is expected to have increasing returns to scale as it represents commitment to high level team dynamics

Empirical Results

The Linear Model

The empiric analysis of our models reveals a complex landscape of salary efficiency, defined by high impact marginal roster spots and diminishing returns of top end Supermax contracts. The linear regression model achieved an Adjusted R^2 of .454, meaning the model explains roughly 45.4% of the variation in NBA win totals. Figure 1 shows our linear models predicted win count vs the realized win count.

Statistical Significance and Magnitude Analysis

The most striking practical takeaway from the linear model is the immense influence of the Allstars and pct45 variables. The Allstars coefficient of 4.4079 (p<0.001) indicates that, holding all other variables constant, the addition of a single All-Star to a roster is worth approximately 4.4 additional wins. This confirm that high level talent is still the primary driver of success.

However, the magnitude of the pct45 coefficient, 47.8213 (p=0.002), reveals a critical strategic insight for roster construction. Practically, this suggests that small percentage-based improvements in the quality of the 4th and 5th starters can be more impactful than the same investment into the top end of a star contract. For an NBA General manager, this suggest that shifting just 2% of salary from a supermax contract can theoretically result in an increase of 0.95 wins.

The Inefficiency of the Supermax Trap

Perhaps the most notable finding is the statistical insignificance of the Supermax variables. While the standard Max variable remains statistically significant (p=0.05) with a coefficient of 1.3794. Indicating that Max contracts in the 25-30% range remains a positive investment with each additional Max player carries an approximate return of additional 1.4 wins. While the rk_supermax (p=0.659) and vt_supermax (p=0.556) variables fail to reach statistical significance. As seen in Figure 2, Max contracts have a positive and linear relationship with wins. Whereas supermax contracts exhibit diminishing marginal returns, eventually leading to a decrease in median wins as the number of top-heavy contracts increases.

From a practical standpoint, this suggests that the addition 5-10% of the salary cap required to retain or acquire a supermax player act as a deadweight loss in terms of marginal wins. While a team may want to retain or acquire a star, the sheer magnitude of the cap hold required prevents them from investing into the rest of the roster, which our models show are significant predictors of team success.

Management and Health as Economic Multipliers

The model also quantifies the value of off-court investments. The Analytics_Staff variable, significant at the 10% level, carrying a coefficient of 0.5039. This suggests that for every two analytics staff hired, a team can expect an average increase of one win per season. Given the relatively low cost of staff compared to player salaries this represents a significant return on investment.

Conversely, the Injured_pct variable has a magnitude of -13.8150 (p=0.028) is a warning to both over allocation into one player and investing into players with extensive injury history or

concerns. Practically, this means that if 10% of the team's salary is lost to games missed due to injury a team should expect to lose approximately 1.38 additional games. This highlights the important of player availability especially when large amounts of capital are concentrated into individuals.

The Quadratic Model

Utilizing Quadratic Regression, we can estimate the ideal range for roster spending. The model only achieved an Adjusted R^2 of .207, meaning the model explained roughly 20.7% of the variation in NBA wins. Despite this, the quadratic model provides potential for practical application as its trends match the ones established in the linear model with estimated ranges for each contract shown in Table 4.

Establishing the Efficiency Frontier

The percentage spent on a team's top player, pct1, has a coefficient of 130.9982 (p=0.249), while its quadratic counterpart has a coefficient of -248.0393 (p=0.344). This means that the relationship of wins and percentage of salary allocated to its top player exhibits an inverted u concave shape, as seen in Figure 3, peaking in efficiency at 26.4%. Using a 99% confidence threshold we can determine an ideal range between 22.5% and 30.2%. Economically, this means that investment into a primary star carries diminishing marginal returns

The percentage spent on a team's second highest paid player, pct2, has a coefficient of -48.9643 (p=0.673), while its quadratic counterpart has a coefficient of 332.4151 (p=0.303). This means that the relationship of wins and percentage of salary allocated to its secondary player exhibits a u convex shape, as seen in Figure 4, having a minimum in efficiency at 7.4%. Using 95% confidence threshold we can determine an ideal range between 27.1% and 28.9%.

Interestingly, the ideal range for a secondary star exceeds the peak of pct1 and sits in the higher end of the ideal range for pct1. Mathematically this seems counter intuitive. However, this phenomenon represents the benefit of investing into a dual star system over a heliocentric system. Economically, this means that investment into a secondary star has increasing returns to scale.

The percentage spent on a team's third highest paid player, pct3, has a coefficient of -99.6096 (p=0.503), while its quadratic counterpart has a coefficient of 462.8646 (p=0.380). This means that the relationship of wins and percentage of salary allocated to its secondary player exhibits a u convex shape, as seen in figure 5, having a minimum in efficiency at 10.8%. Using a 95% confidence threshold we can determine an ideal range between 19.8% and 22.0%.

Similarly to the relationship between pct2 and pct1, the minimum point of efficiency at 10.8% is actually higher than the minimum point of efficiency in the pct2 partial effect, appearing counter intuitive. This illustrates the lower margin of error when it comes to tertiary stars as well as the appeal of the "Big Three" roster construction strategy seen in the NBA. Economically, this means that investment into a tertiary star has increasing returns to scale.

The relationship between percent spent on a team's fourth high paid player (pct4) differs from pct1, pct2, and pct 3. The relationship exhibited is linear, instead of quadratic with pct 4 having a coefficient of 76.6171 and significance at the 10% level (p=0.059). This means that for every 1% of additional salary allocated to your 4th highest paid player you can anticipate approximately .77 additional wins.

While the linear nature of pct 4 makes predicting an ideal range more difficult than a quadratic. We solve this by creating a subset of the data for teams that exceeded 50 wins and

made our range within one standard deviation from the mean values of those teams. This approach gave the ideal range as between 7.5% and 12.6%, as seen in Figure 6.

Similarly, the percentage spent on a team's 5th highest paid player follows a linear relationship. With a coefficient of 123.6491 and the highest significance (p=0.006), as seen in Figure 7, pct5 clearly illustrates the relationship between quality depth and wins in the NBA. Meaning that for every additional 1% allocated to a team's 5th highest paid player an approximate return of 1.24 wins can be expected, all else held equal. Using the same methodology for estimating the ideal range for pct4, we estimate the ideal range for investment in pct5 as between 5.3% and 9.0%.

Conclusion

The 2017 NBA collective bargaining agreement fundamentally changed how the league operates; team success is increasingly a product of roster optimization rather than mere talent accumulation. While the presence of elite talent remains a foundational requirement for competitiveness, the institutional constraints imposed by the Supermax contract classification has created a diminishing returns trap for franchises. The primary finding of this study is that the marginal utility of a 35% cap hold, regardless of the player pedigree, is significantly lower than the value generated by a high-quality 5th starter (pct5) and aggregate roster depth (pct45).

The practical implications for NBA front offices are twofold. First, the statistical significance of Analytics_Staff suggests that the "Moneyball Effect" has shifted from player identification to player allocation. In an era with strictly enforced salary tiers, the most successful teams will be those that utilize data to identify players that provide a surplus of value to fill the 4th and 5th roster spots within the ideal spending ranges of 5.3% to 12.6%. Second, the

insignificance of the rk_supermax and vt_supermax variables suggest that GMs should exercise extreme caution when triggering performance-based escalators, as the resulting cap illiquidity often forces teams into a top-heavy structure that the quadratic model identifies as inefficient.

However, several qualifications must be addressed. The adjusted R^2 values (0.454 for linear and 0.207 for quadratic) suggests that while salary allocation is a critical driver of wins, it does not explain everything. Factors such as coaching quality, team chemistry and clutch performance are difficult to quantify and likely account for a large portion of the remaining variance. In addition, this study is limited by the specific period of study, 2017-2026, a shift in the next collective bargaining agreement could fundamentally alter the Efficiency Frontier identified here.

Another Consideration, this study intentionally excludes any player specific skill analysis. While optimal player compositions depend heavily on team context, this paper instead focuses on how teams allocate financial resources across the roster. Treating salary as a proxy for expected contribution. This allows for a more generalized analysis of roster construction under salary cap constraints.

Further research should focus on the interaction between analytics investment and the supermax. Specifically, can a high functioning analytics department mitigate the negative impact of a 35% cap hold by consistently finding minimum scale players who produce at starter level efficiency. In addition, evaluating whether the identified trends continue beyond the regular season and into the playoffs. As the NBA continues to lean into data driven decision making, the intersection of contract law and economic modeling will remain among the most fertile ground for achieving a competitive advantage.

Tables and Figures

Table 1: *Summary Statistics*

index	Wins	pct1	pct2	pct3	pct4	pct5	Wins (t-1)	Time Period	All Stars	Analytics Staff	Injured pct	Max	Rk super max	Vt super max	Pct 45
count	270	270	270	270	270	270	270	270	270	270	270	270	270	270	270
mean	39.77	0.22	0.17	0.12	0.09	0.07	39.7	5	0.86	3.88	0.19	0.89	0.04	0.28	0.16
std	11.89	0.04	0.04	0.04	0.02	0.02	11.96	2.59	0.78	1.97	0.09	0.84	0.198	0.57	0.04
min	15	0.06	0.04	0.04	0.02	0.02	10	1	0	0	0.04	0	0	0	0.04
25%	31	0.19	0.14	0.09	0.07	0.05	32	3	0	2.5	0.13	0	0	0	0.13
50%	41.5	0.22	0.17	0.12	0.09	0.07	41	5	1	4	0.18	1	0	0	0.16
75%	48.75	0.25	0.2	0.14	0.11	0.08	48	7	1	5	0.24	1	0	0	0.19
max	68	0.34	0.29	0.22	0.17	0.12	73	9	3	12	0.49	3	1	3	0.27

Table 2: *Linear Regression*

Variables	Coefficients	STD error	P-Value
Intercept	12.3624	3.213	0.000
Wins_t_minus_1	0.3859	0.052	0.000
Time_Period	-0.0107	0.238	0.964
AllStars	4.4079	0.851	0.000
Injured_pct	-13.8150	6.260	0.028
Analytics_Staff	0.5039	0.303	0.098
Max	1.3794	0.701	0.050
rk_supermax	1.2879	2.917	0.659
vt_supermax	0.6719	1.141	0.556
pct45	47.8213	15.404	0.002
R^2	**.473**	**Adj R^2**	**.454**

Table 3: *Quadratic Regression*

Variables	Coefficients	STD error	P-Value
Intercept	10.8200	11.040	0.328
pct1	130.9982	113.348	0.249
pct1^2	-248.0393	261.596	0.344
pct2	-48.9643	115.765	0.673
pct2^2	332.4151	321.828	0.303
pct3	-99.6096	148.675	0.503
pct3^2	462.8646	526.488	0.380
pct4	76.6171	40.463	0.059
pct5	123.6491	44.543	0.006
R^2	**.230**	**Adj R^2**	**.207**

Table 4: *Formula for Success*

Player	Ideal Range	Min/Max	Takeaway
1st	22.5% - 30.2%	Max: 26.4%	Efficiency peaks at 26.4% spending beyond 30% creates a “Supermax Trap”
2nd	27.1% - 28.9%	Min: 7.4%	Requires high level spending to increase returns to wins
3rd	19.8% - 22.0%	Min: 10.8%	High win elasticity
4th	7.5% - 12.6%	–	Glue guys where significant wins increase
5th	5.3% - 9.0%	–	Most statistically significant creates a high floor

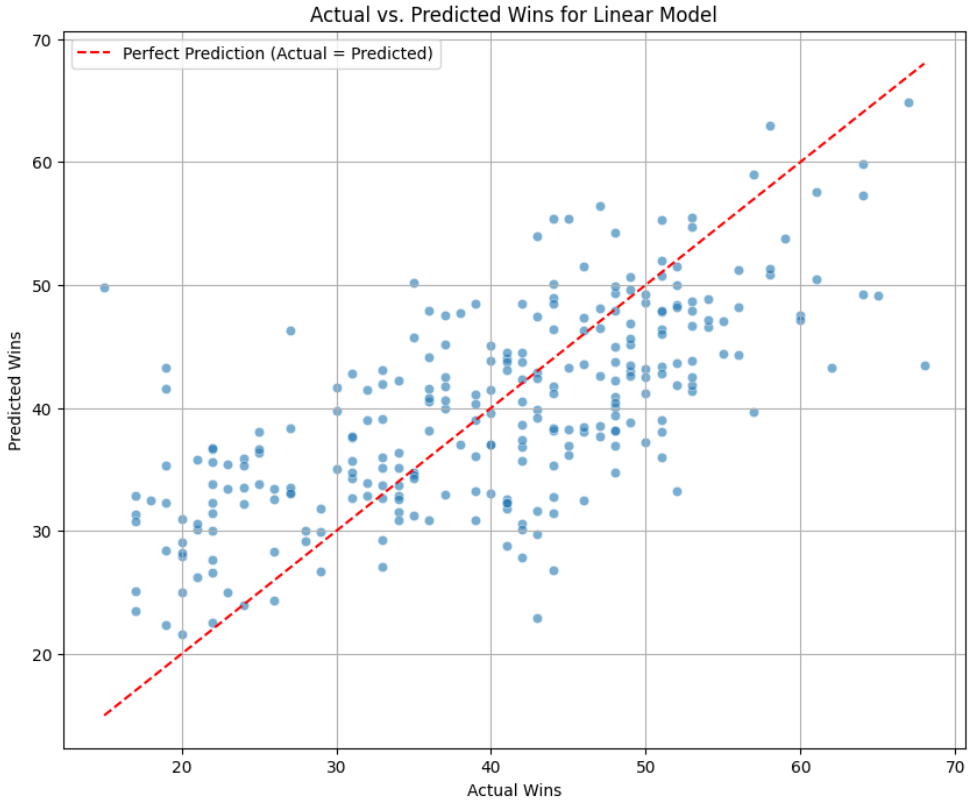


Figure 1: *Linear Model Actual vs Predicted Wins*

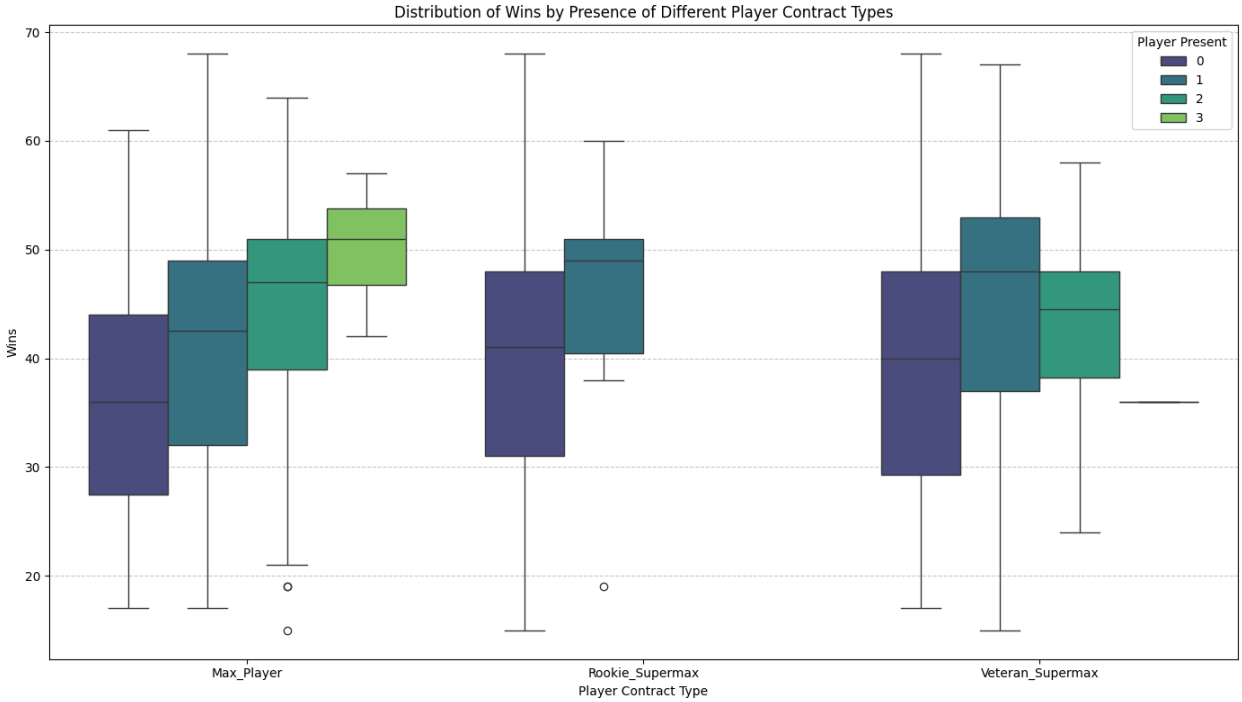


Figure 2: *Distribution of Wins by Contract Types*

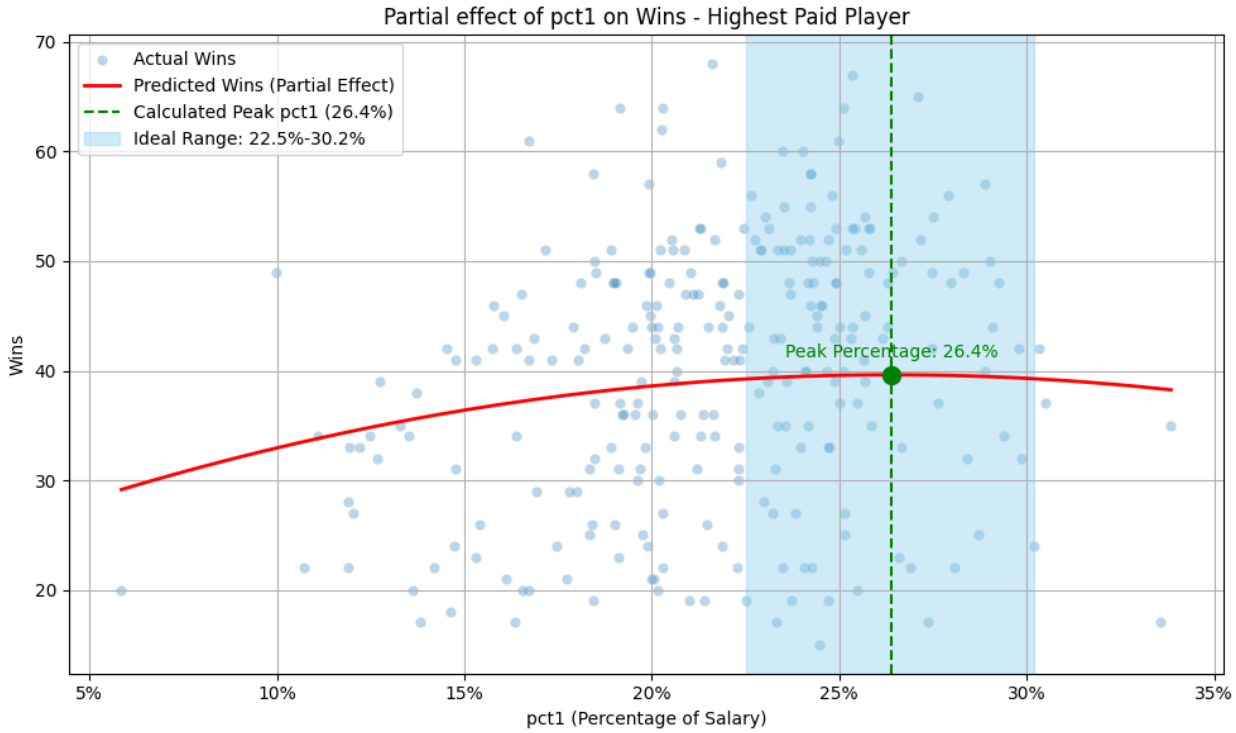


Figure 3: *Partial Effect of pct1 on Wins*

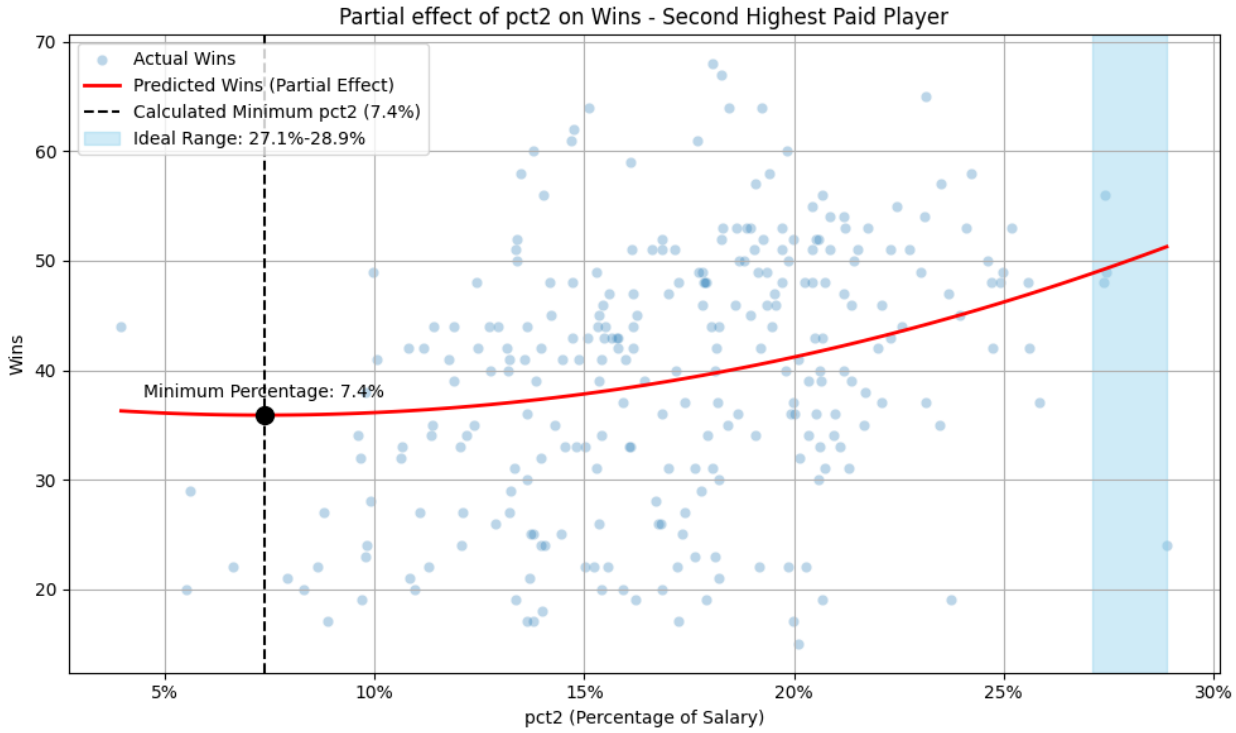


Figure 4: *Partial Effect of pct2 on Wins*

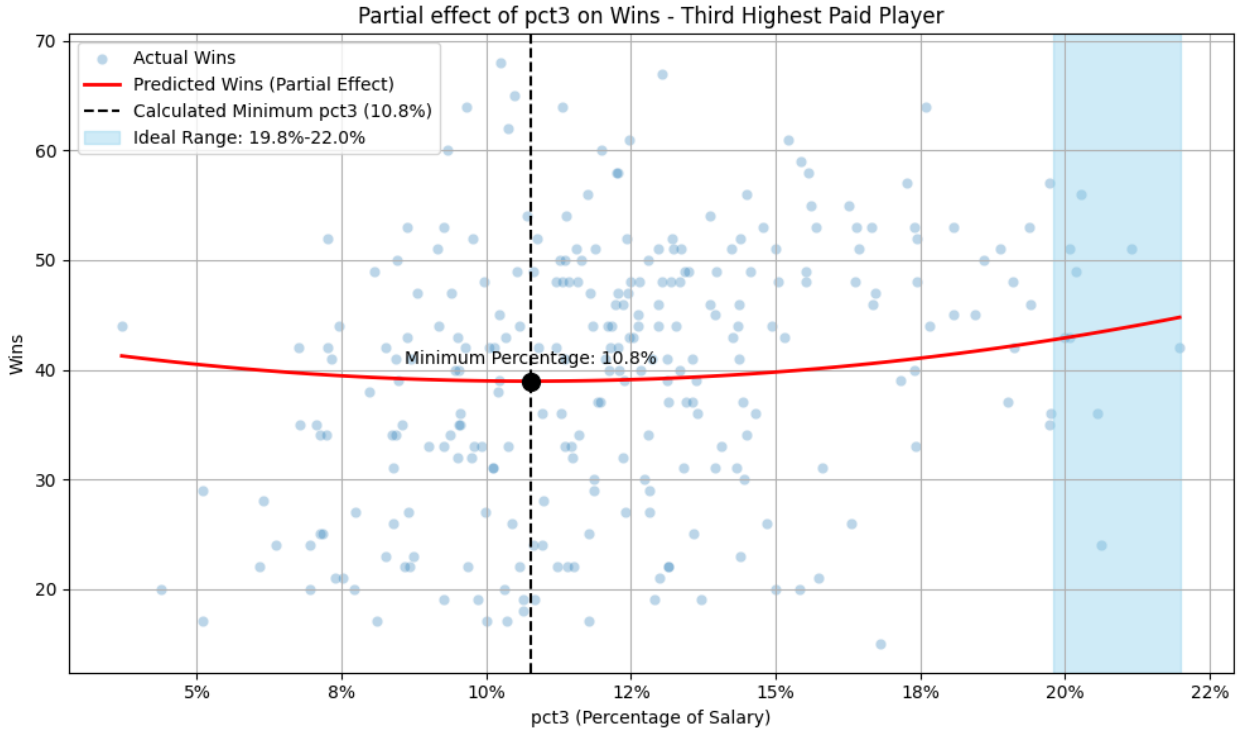


Figure 5: *Partial Effect of pct3 on Wins*

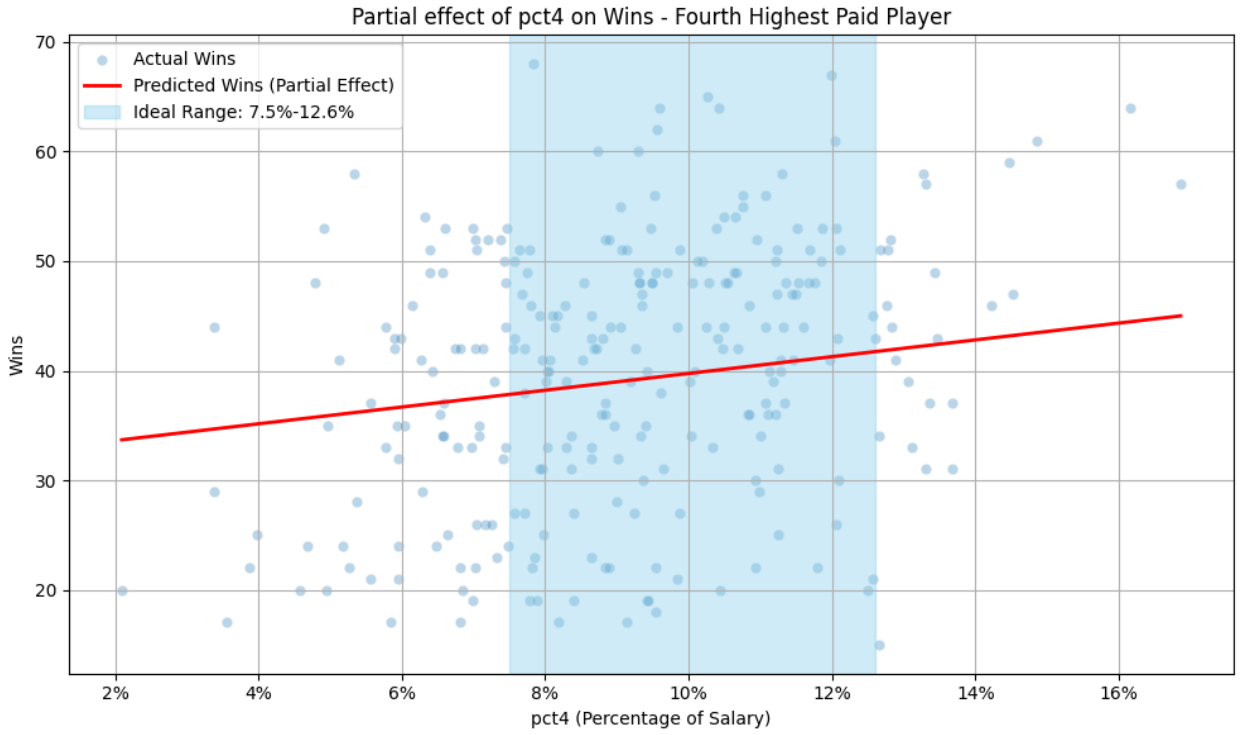


Figure 6: *Partial Effect of pct4 on Wins*

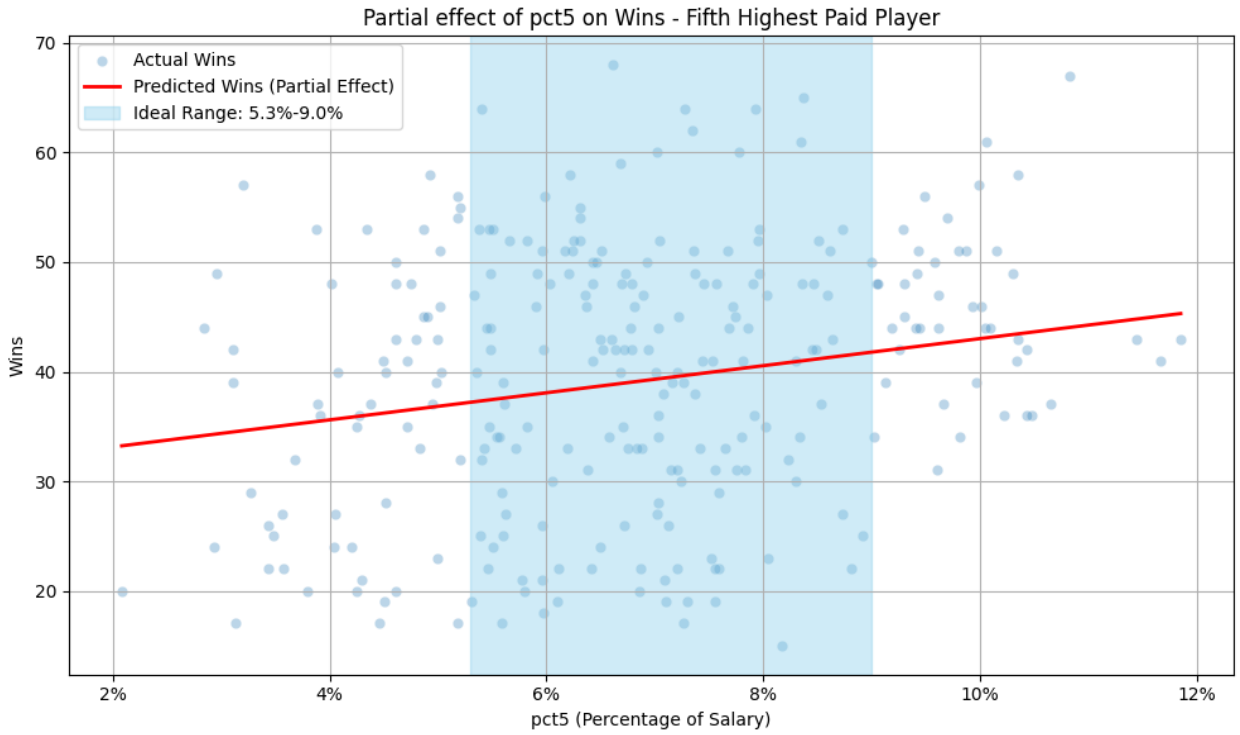


Figure 7: *Partial Effect of pct5 on Wins*

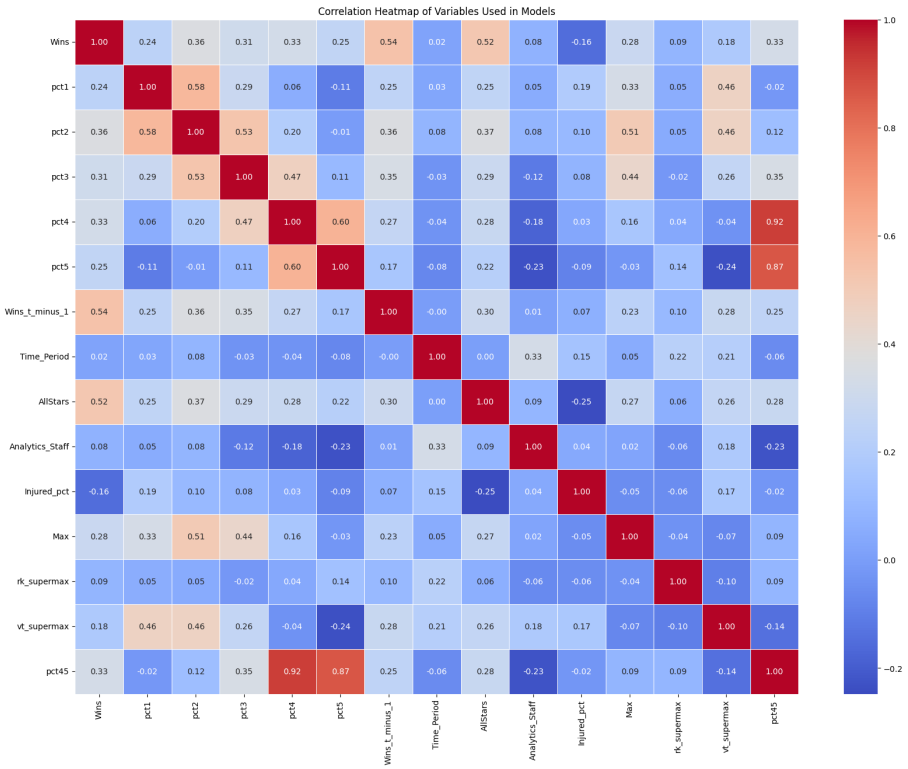


Figure 8: *Correlation Heat Map*

References

Cynkar, Anderson. “Optimal Capital Allocation Decisions and Investments in the NBA .” Scholarships @ Claremont, 2025. https://scholarship.claremont.edu/cgi/viewcontent.cgi?article=5112&context=cmc_theses.

NBAstuffer. “NBA Analytics Departments: Team-by-Team Staff List.” NBAstuffer, April 13, 2026. https://www.nbastuffer.com/analytics101/nba-teams-that-have-analytics-department/.

RealGM. 2025-2026 NBA all-star game rosters. Accessed April 12, 2026. https://basketball.realgm.com/nba/allstar/game/rosters/2026.

Spotrac. “2025 NBA Cap Hit Rankings.” spotrac.com, 2026. https://www.spotrac.com/nba/rankings/player/_/year/2025/sort/cap_total.

Spotrac. “2025 NBA Cap% of League Cap Rankings.” spotrac.com, 2026. https://www.spotrac.com/nba/rankings/player/_/year/2025/sort/cap_total_league_pct.

Spotrac. “2025-26 NBA Team Salary Cap Tracker.” spotrac.com, 2026. https://www.spotrac.com/nba/cap/_/year/2025/sort/cap_total.

Wang, Henry, and Arnab Sarker. “The Effect of Basketball Analytics Investment on National Basketball Association (NBA) Team Performance.” Sage Journals, 2025. https://journals.sagepub.com/doi/10.1177/15270025251328264#tab-contributors.

“Wayback Machine.” Expand web menu. Accessed April 1, 2026. https://web.archive.org/.